

Experiment ‘p_4_4_month_with_highest_temperature_seq’

Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing assume : 0
Responses containing expect : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_4_4_month_with_highest_temperature_seq

Natural language statement: Write a method that reads a sequence of twelve temperature values and returns the number of the month with the highest temperature.

Method signature: p_4_4_month_with_highest_temperature_seq(inputs: seq<real>) returns (hottest: int)

Ensures

- $1 \leq \text{hottest} \leq 12$
- forall $i :: 0 \leq i \leq |\text{inputs}| - 1 \implies \text{inputs}[\text{hottest} - 1] \geq \text{inputs}[i]$

Requires

- $|\text{inputs}| = 12$

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that reads a sequence of twelve temperature values and returns the number
→ of the month with the highest temperature.

The signature should be:

p_4_4_month_with_highest_temperature_seq(inputs: seq<real>) returns (hottest: int)

The method should respect the following contract:

requires |inputs| == 12, ensures 1 <= hottest <= 12, ensures forall i :: 0 <= i <= |
→ inputs| - 1 ==> inputs[hottest - 1] >= inputs[i]

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmfWsvOYrs1eUUDGyZbTmY0Dv2Gsr

Model: gpt-5.5-2026-04-23

Created at: 1780491754

Finish reason: stop

Usage: 358 tokens in, and 654 tokens out

```
{
  var maxIdx := 0;
  var i := 1;
  while i < |inputs|
    invariant 1 <= i <= |inputs|
    invariant 0 <= maxIdx < i
    invariant forall j :: 0 <= j < i ==> inputs[maxIdx] >= inputs[j]
    decreases |inputs| - i
  {
    if inputs[i] > inputs[maxIdx] {
      maxIdx := i;
    }
    i := i + 1;
  }
  hottest := maxIdx + 1;
}
```

Final Program

```
method p_4_4_month_with_highest_temperature_seq(inputs: seq<real>) returns (hottest: int)
  requires |inputs| == 12
  ensures 1 <= hottest <= 12
  ensures forall i :: 0 <= i <= |inputs| - 1 ==> inputs[hottest - 1] >= inputs[i]
{
  var maxIdx := 0;
  var i := 1;
  while i < |inputs|
    invariant 1 <= i <= |inputs|
    invariant 0 <= maxIdx < i
    invariant forall j :: 0 <= j < i ==> inputs[maxIdx] >= inputs[j]
    decreases |inputs| - i
  {
    if inputs[i] > inputs[maxIdx] {
      maxIdx := i;
    }
    i := i + 1;
  }
  hottest := maxIdx + 1;
}
```

Total Token Usage

Input tokens: 358

Output tokens: 654

Reasoning tokens: 512

Sum of 'total tokens': 1012

Experiment Timings

Overall Experiment started at 1780491754217, ended at 1780491772800, lasting 18583ms (18.58 seconds)

Iteration #1 started at 1780491754218, ended at 1780491772800, lasting 18582ms (18.58 seconds)